8.4.36

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Question

Two sets A and B are as under:
$A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\}$
$B = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\}$

1. $A \subset B$
2. $A \cap B$
3. neither $A \subset B$ nor $B \subset A$
4. $B \subset A$

Theoretical Solution

Set A:

$$A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\} \tag{1}$$

$$|a - 5| < 1 \implies 4 < a < 6 \text{ and } |b - 5| < 1 \implies 4 < b < 6 \tag{2}$$

This describes the interior of a square in the ab-plane. The square is centered at $(5, 5)$ and its sides are the lines $a = 4$, $a = 6$, $b = 4$, and $b = 6$.

Theoretical Solution

Set B:

$$B = \left\{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a-6)^2 + 9(b-5)^2 \leq 36\right\} \tag{3}$$

$$\frac{(a-6)^2}{3^2} + \frac{(b-5)^2}{2^2} \leq 1 \tag{4}$$

This can be expressed in the form ,

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f \leq 0$$

where,

$$\mathbf{V} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -24 \\ -45 \end{pmatrix} \text{ and } f = 333. \tag{5}$$

This describes the region of an ellipse, including its interior, in the ab-plane. The ellipse is centered at (6,5) and has a horizontal semi-major axis of length 3 and a vertical semi-minor axis of length 2.

Theoretical Solution

Now, we substitute the maximum possible values for these terms into the expression for the ellipse to see if the condition for Set B is met:

On substitutting $\mathbf{x} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$,

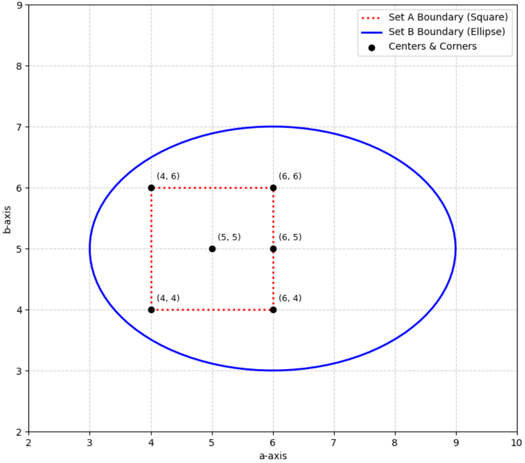
$$\begin{pmatrix} 4 & 6 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} -24 & -45 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \end{pmatrix} + 333 \tag{6}$$

$$\implies -11 \leq 0 \tag{7}$$

Therefore, every point in Set A is contained within Set B.
This means that A is a subset of B.

Therefore, $A \subset B$

Plot


Set A Boundary (Square)
Set B Boundary (Ellipse)
Centers & Corners
(4, 6)
(6, 6)
(5, 5)
(6, 5)
(4, 4)
(6, 4)
a-axis
b-axis
2
3
4
5
6
7
8
9
10